Class 19 Learning Activity

Reaction Engineering Basics, The Course

In concentrated aqueous solutions, the irreversible condensation reaction (1) is highly exothermic, $\Delta H_1 = -184$ kJ mol^{-1}. The rate expression is given in equation (2) where the pre–exponential factor equals 4.866 x 10^{11} m^3 mol^{-1} min^{-1} and the activation energy equals 74.1 kJ mol^{-1}. A stirred tank reactor operating at atmospheric pressure will be charged with 4.0 L of a solution containing A at a concentration of 10 mol L^{-1} and 20°C. The reactor is cooled using water at 20 °C flowing into a perfectly mixed, 500 cm^3 shell with a heat transfer area of 550 cm^2 and a heat transfer coefficient of 8480 J m^{-2} min^{-1} K^{-1}. The cooling water flows at a steady rate of 1.0 L min^{-1}, and initially the water in the shell is at 20 °C. The reacting fluid and the cooling water may be assumed to have a heat capacity and density equal to that of water, 1 cal g^{-1} K^{-1} and 1 g cm^{-3}, respectively. A 10 M solution of B at 20 °C will be added to the reacting fluid at a constant flow rate. What is the maximum volumetric flow rate at which the base solution at 20 °C can be added to the reactor without causing the reacting fluid temperature to exceed 80 °C?

$$A + B \rightarrow 2Z \tag{1}$$

$$r_1 = k_1 C_A C_B \tag{2}$$